Replicating and stress-testing stock prediction from Reddit WallStreetBets posts

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1. Problem statement

Replicate the CS229 report 'Predicting market volatility and building short-term trading strategies using data from Reddit's WallStreetBets', then build and compare further models and a more rigorous evaluation. Task: from a WSB post that mentions GME, AMC, TSLA or FB, predict (a) whether the stock's open rises at the next trading day (direction) and (b) whether the move is large (volatility).

2. Dataset

Kaggle 'gpreda/reddit-wallstreetsbets-posts' (title, score, comments, body, timestamp) joined with Yahoo Finance daily open prices. Posts mentioning GME, AMC, TSLA or FB give AMC 5,336, FB 260, GME 14,688, TSLA 1,238 labelled posts. Features (paper): post score, comment count, length, VADER positive/neutral/negative sentiment (+ bias). Labels: direction = open(t1) > open(t0), volatility = |open(t1)/open(t0)-1| above the training median, where t0 is the first trading day on/after the post and t1 the next trading day.

3. Approach

(1) Replicate: Newton's-method logistic regression written from scratch and a 3x256 ReLU network (Adam 1e-3, batch 16, 30 epochs), plus random/majority baselines. (2) Extend: six extra model families (NN on extended features, tuned logistic regression, Random Forest, Gradient Boosting, SVM-RBF, KNN) and extra features (log counts, caps ratio, hour, weekday, daily mentions, previous return, 5-day volatility, TF-IDF+SVD). (3) Evaluate rigorously: three 80/10/10 splits - paper-style random post shuffle (leaky), day-grouped, and chronological with an embargo day; leakage-free tuning with day-based CV folds; neural nets averaged over 5 seeds; balanced accuracy, F1, AUC and bootstrap CIs; walk-forward long/flat backtest with costs, block bootstrap and a permutation test.

4. Implementation overview

Python (pandas, scikit-learn, NumPy). src/pipeline.py (cleaning, VADER, labels, day-based splits/CV), src/models.py (NewtonLR, SeedAvgNN, model roster and search spaces), src/run.py (experiments, tables, plots), src/backtest.py (walk-forward backtest), tests/ (unit tests), run_all.sh (one-command reproduction).

5. Results

Table 1. Mean over the four stocks. Accuracy on the random (leaky), day-grouped and chronological test sets, and balanced accuracy on the day-grouped set. First rows are baselines, then the paper's models, then ablations and new models.

Model	Dir rand	Dir grp	Dir chr	Vol rand	Vol grp	Vol chr	BalAcc dir	BalAcc vol
Majority baseline	0.652	0.589	0.542	0.650	0.837	0.890	0.500	0.500
Random baseline	0.482	0.484	0.429	0.506	0.410	0.549	0.482	0.410
LogReg Newton (paper)	0.658	0.563	0.516	0.614	0.723	0.701	0.512	0.653
NN 3x256 (paper)	0.647	0.453	0.461	0.613	0.607	0.612	0.473	0.575
RF, paper feats	0.671	0.462	0.508	0.657	0.567	0.577	0.486	0.576
HGB, paper feats	0.664	0.426	0.513	0.647	0.491	0.594	0.457	0.495
NN 3x256, ext feats	0.798	0.475	0.491	0.816	0.695	0.640	0.476	0.724
LogReg, tuned L2	0.693	0.513	0.535	0.758	0.609	0.647	0.485	0.553
RandomForest	0.880	0.586	0.503	0.911	0.835	0.724	0.601	0.771
HistGradBoost	0.868	0.536	0.497	0.905	0.720	0.573	0.571	0.698
SVM-RBF	0.757	0.439	0.502	0.774	0.755	0.696	0.458	0.750
KNN	0.755	0.461	0.477	0.769	0.668	0.675	0.474	0.643

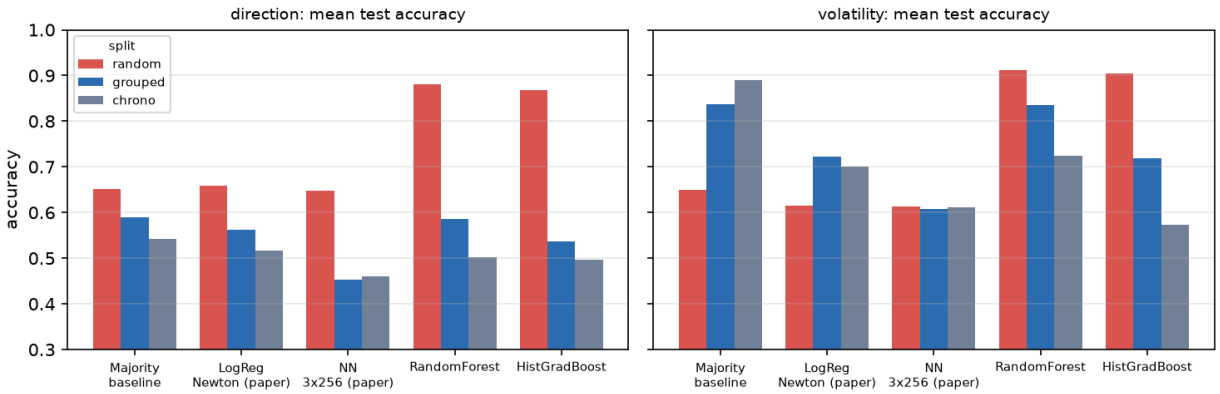


Figure 1. Accuracy drops once the leak (same-day posts across train and test) is removed.

Table 2. Walk-forward long/flat backtest (direction task, retrained every 10 days, 10 bps cost per entry/exit, same-day buy-and-hold as benchmark; p-value from a permutation test of the timing).

Stock	Model	Days	Exposure	Return	Buy&hold	Sharpe	Perm. p
GME	LogReg-Newton	61	0.00	+0.0%	-9.7%	-	1.00
GME	RandomForest	61	0.23	+10.7%	-9.7%	1.77	0.29
GME	HistGradBoost	61	0.38	+17.2%	-9.7%	1.19	0.17
AMC	LogReg-Newton	61	0.51	+34.9%	+168.7%	1.56	0.65
AMC	RandomForest	61	0.56	+1.7%	+168.7%	0.58	0.88
AMC	HistGradBoost	61	0.56	-35.1%	+168.7%	-0.91	0.98
TSLA	LogReg-Newton	62	0.00	+0.0%	-13.2%	-	1.00
TSLA	RandomForest	62	0.06	+0.2%	-13.2%	0.11	0.40
TSLA	HistGradBoost	62	0.32	-13.1%	-13.2%	-2.27	0.77
FB	LogReg-Newton	26	1.00	+12.9%	+12.9%	5.67	1.00
FB	RandomForest	26	0.92	+12.2%	+12.9%	5.33	0.36
FB	HistGradBoost	26	0.88	+11.1%	+12.9%	4.89	0.46

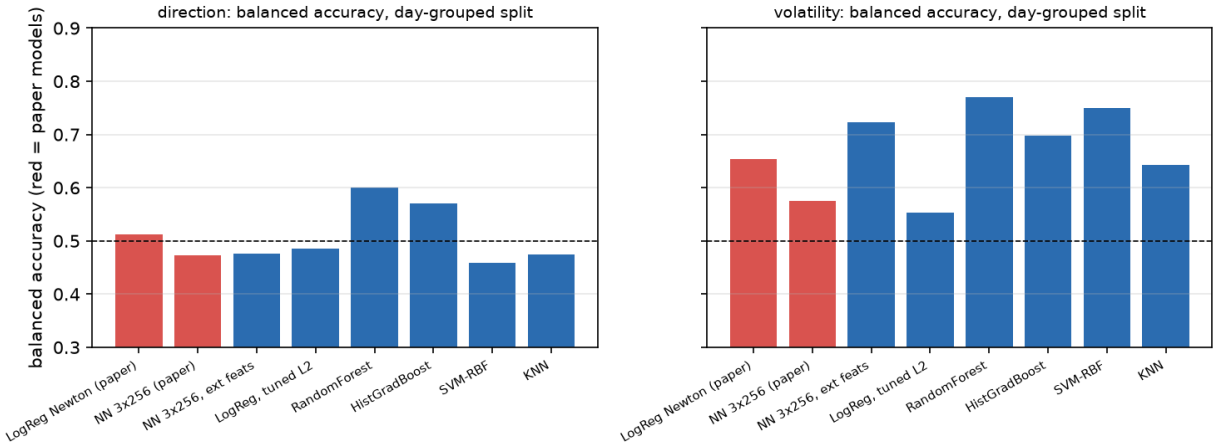


Figure 2. Balanced accuracy on the day-grouped split (red = paper models, dashed = chance).

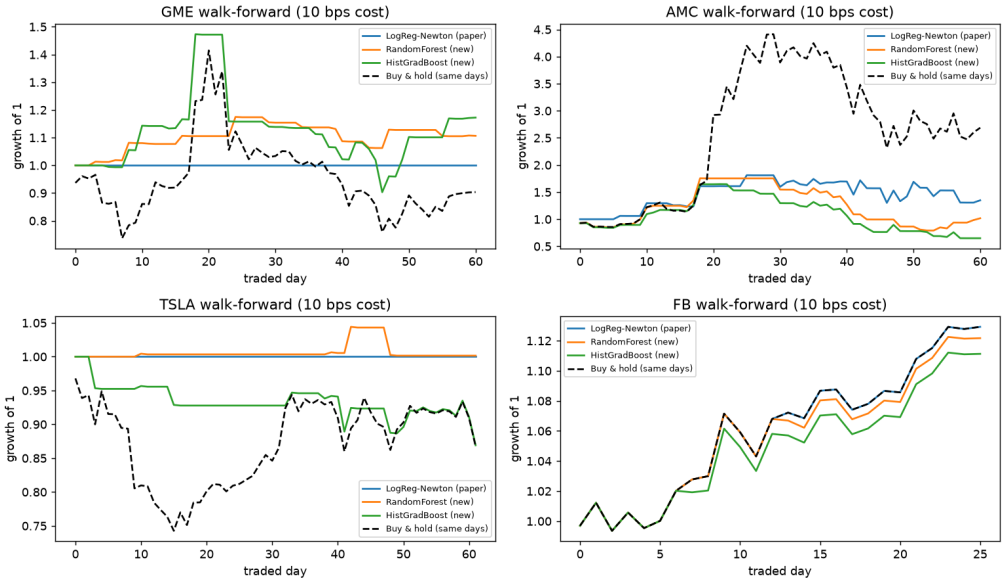


Figure 3. Equity curves of the walk-forward backtest.

6. Conclusions

• Leakage: the paper-style random split lets same-day posts (which share one label) fall in both train and test. Random Forest direction accuracy falls from 0.880 (random) to 0.586 (day-grouped); volatility from 0.911 to 0.835. The paper's own models barely move, so reported paper-style accuracies are likely optimistic.

• Direction, leak-free: best new model by balanced accuracy is RandomForest (0.601) vs best paper model LogReg Newton (paper) (0.512); however the best new raw accuracy 0.586 does not beat the majority baseline (0.589).

• Volatility, leak-free: best new model by balanced accuracy is RandomForest (0.771) vs best paper model LogReg Newton (paper) (0.653); however the best new raw accuracy 0.835 does not beat the majority baseline (0.837).

• Walk-forward backtest (about 61 traded days per stock, 10 bps costs): 0 of 12 stock-model combinations beat random timing at $p<0.05$ (smallest $p = 0.17$), so there is no statistically reliable trading edge.

• Limitations: few days per stock, tiny chronological test sets, regime shifts around the Jan 2021 spike; results are exploratory. Chronological-split numbers should be read with caution.